

# “Retail Pulse” End to End Sales Analytics

## Tools Used:

- MySQL — Data Storage & Querying
- Excel — Quick Analysis & Pivot Reports
- Python (Pandas, NumPy, Matplotlib, Seaborn) — Deep Analysis
- Power BI — Interactive Dashboard



# Problem Statement

A mid-sized retail chain operating across multiple cities in India is experiencing declining sales in certain regions.

## Dataset: RetailPulse Sales Data

- Sales — 1,500 rows — Core transaction data
- Customers — 200 rows — Customer demographics
- Products — 50 rows — Product catalogue
- Regions — 5 rows — City & zone details
- Sales Reps — 20 rows — Rep & region mapping

**Data Period:** January 2022 — December 2023



# Key Findings from MySQL Analysis (Filtered: Delivered orders only)

- **Top Region: Bengaluru** generated the highest revenue of ₹38,327 followed by **Chennai** at ₹37,700
- **Lowest Region: Mumbai** with ₹22,741 — significantly lower than other regions
- **Top Product: Camera** was the best selling product with ₹15,416 in revenue
- **Peak Month: July** recorded the highest monthly revenue of ₹19,304 while **March** was the lowest at ₹10,019
- **Top Segment:** Corporate led with 231 orders and ₹45,369 in revenue
- **Top Sales Rep:** Anand Venkat led with ₹20,001 in revenue
- **Top Category:** Electronics dominates at ₹71,110 followed by Clothing at ₹50,291
- **Top Category-Region Combo:** Electronics in Delhi at ₹17,659

| region_name | revenue  |
|-------------|----------|
| Bengaluru   | 38327.61 |
| Chennai     | 37700.58 |
| Kolkata     | 35626.47 |
| Delhi       | 34716.98 |
| Mumbai      | 22741.97 |

| product_name | revenue  |
|--------------|----------|
| Camera       | 15416.64 |
| Tablet       | 9641.05  |
| Speaker      | 9606.31  |
| Mobile Phone | 8655.75  |
| Ethnic Wear  | 6860.39  |
| Jacket       | 6505.45  |
| Laptop       | 6443.36  |
| Men's Shirt  | 6283.99  |
| Trousers     | 6021.94  |
| Sportswear   | 5576.79  |

| rep_name         | revenue  |
|------------------|----------|
| Anand Venkat     | 20001.61 |
| Bhavana Krishnan | 18308.81 |
| Deepak Natarajan | 17377.86 |

| category       | revenue  |
|----------------|----------|
| Electronics    | 71110.09 |
| Clothing       | 50291.57 |
| Home & Kitchen | 23627.36 |
| Beauty         | 15866.7  |
| Grocery        | 8217.89  |

```
select region_name, category, round(sum(total_amount),2) as revenue
from products
join sales
on products.product_id = sales. product_id
join regions
on regions.region_id = sales.region_id
where order_status = 'Delivered'
group by region_name, category
order by 3 desc;
```

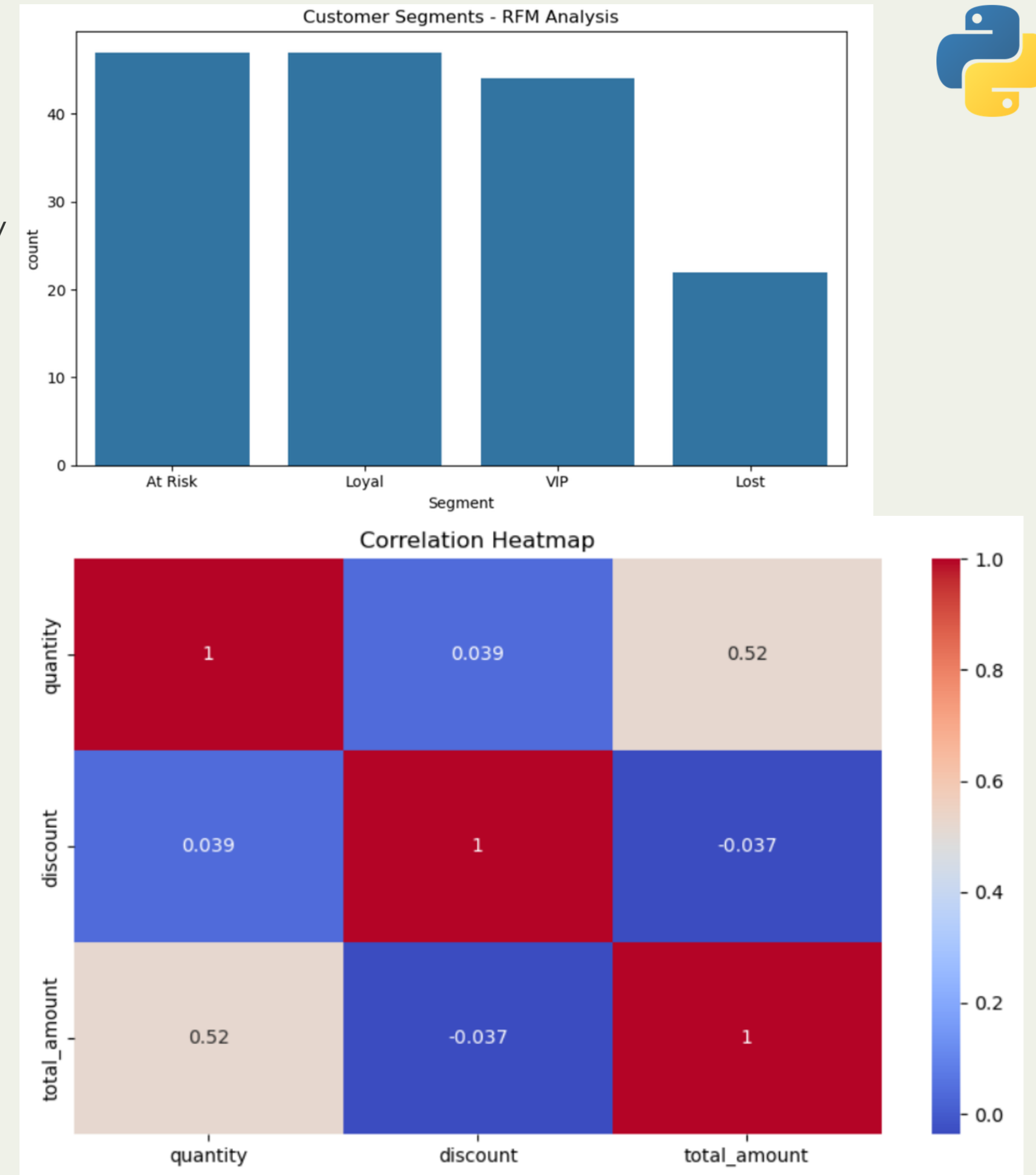
# Key Findings from Python Analysis

## RFM Customer Segmentation:

- **VIP Customers: 44** - high value, buy recently and frequently
- **Loyal Customers: 47** - consistent buyers worth retaining
- **At Risk Customers: 47** - used to buy but going inactive
- **Lost Customers: 22** - lowest engagement, need re-activation campaign

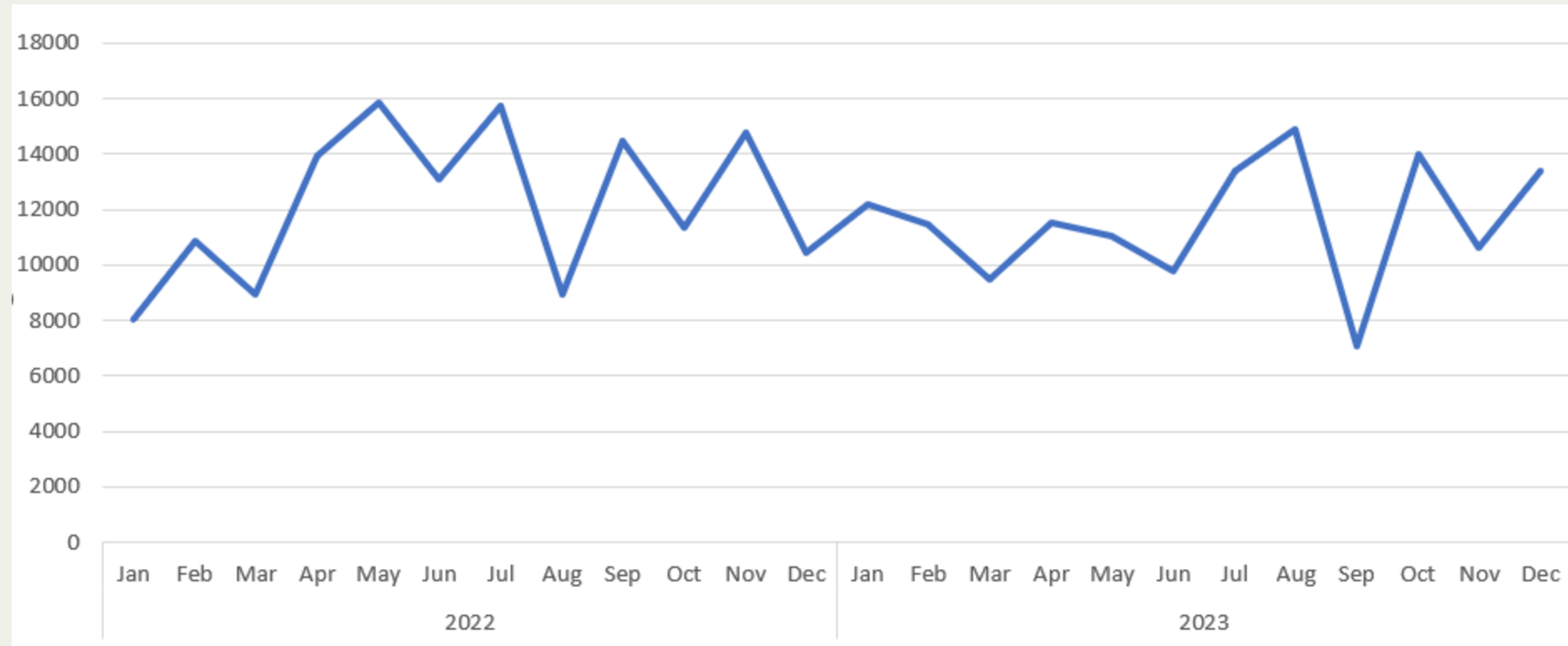
## Correlation Analysis:

- **Quantity vs Revenue = 0.52** - moderate positive correlation, more units sold means more revenue
- **Discount vs Revenue = -0.037** - nearly zero correlation, discounts have no impact on revenue
- **Key Finding:** Discount strategy is ineffective - confirmed by both Excel and Python analysis





# Key Findings from Excel Analysis - Monthly Revenue Trend — 2022 vs 2023

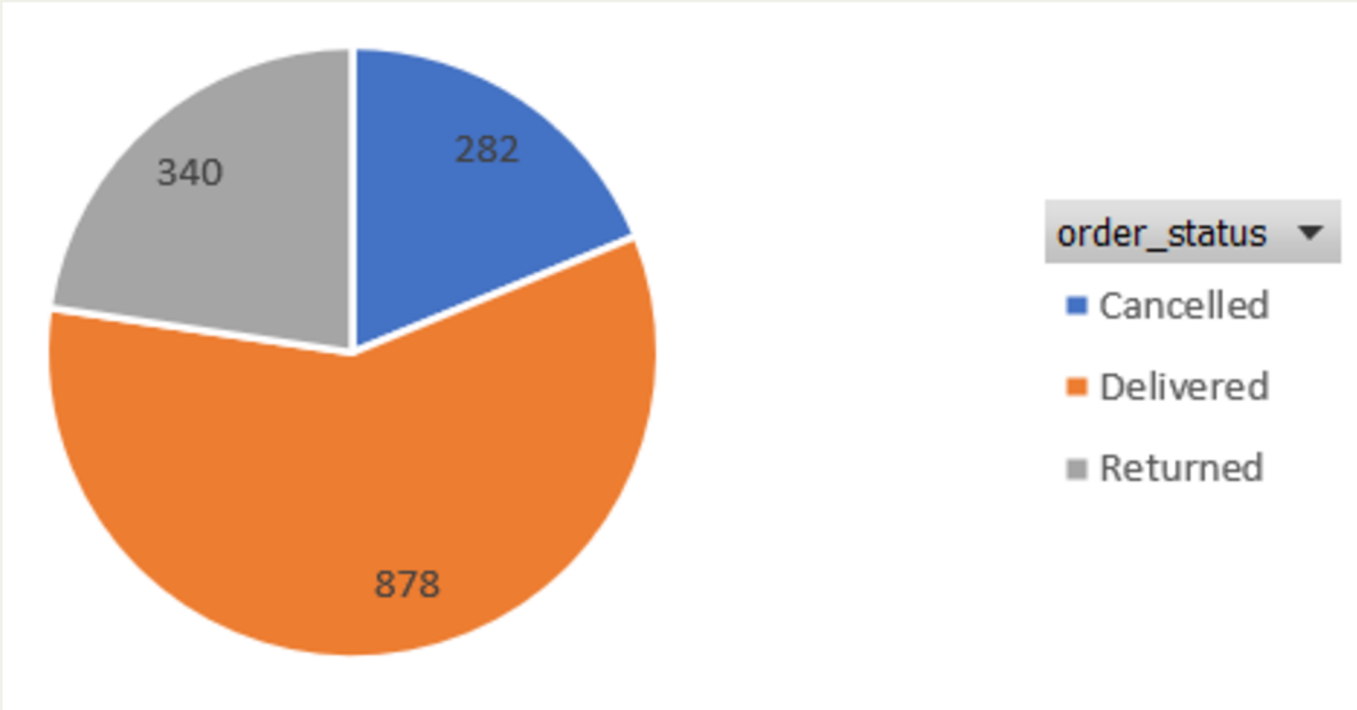


**Overall Trend:**  
Revenue is highly inconsistent across 2022 and 2023 with **no clear growth pattern**

- **2022 Peak:** Strong peak in **May** and **July** 2022 reaching **₹16,000**
- **2022 Dip:** Sharp decline in **August 2022** dropping to **₹9,000**
- **2023 Peak:** Revenue recovered in **July** and **August** 2023 reaching **₹15,000**
- **2023 Low:** Severe drop in **September 2023** hitting the lowest point at **₹7,000**

**Key Concern:** 2023 overall revenue is visibly lower than 2022, directly confirming the declining sales problem stated in our problem statement

# Key Findings from Excel Analysis - Order Status & Discount Analysis



- **Order Status Split:** Out of 1,500 orders — 878 Delivered (58.5%), 340 Returned (22.7%), 282 Cancelled (18.8%)
- **Key Concern:** Nearly **41%** of orders **never completed successfully**, a major revenue loss

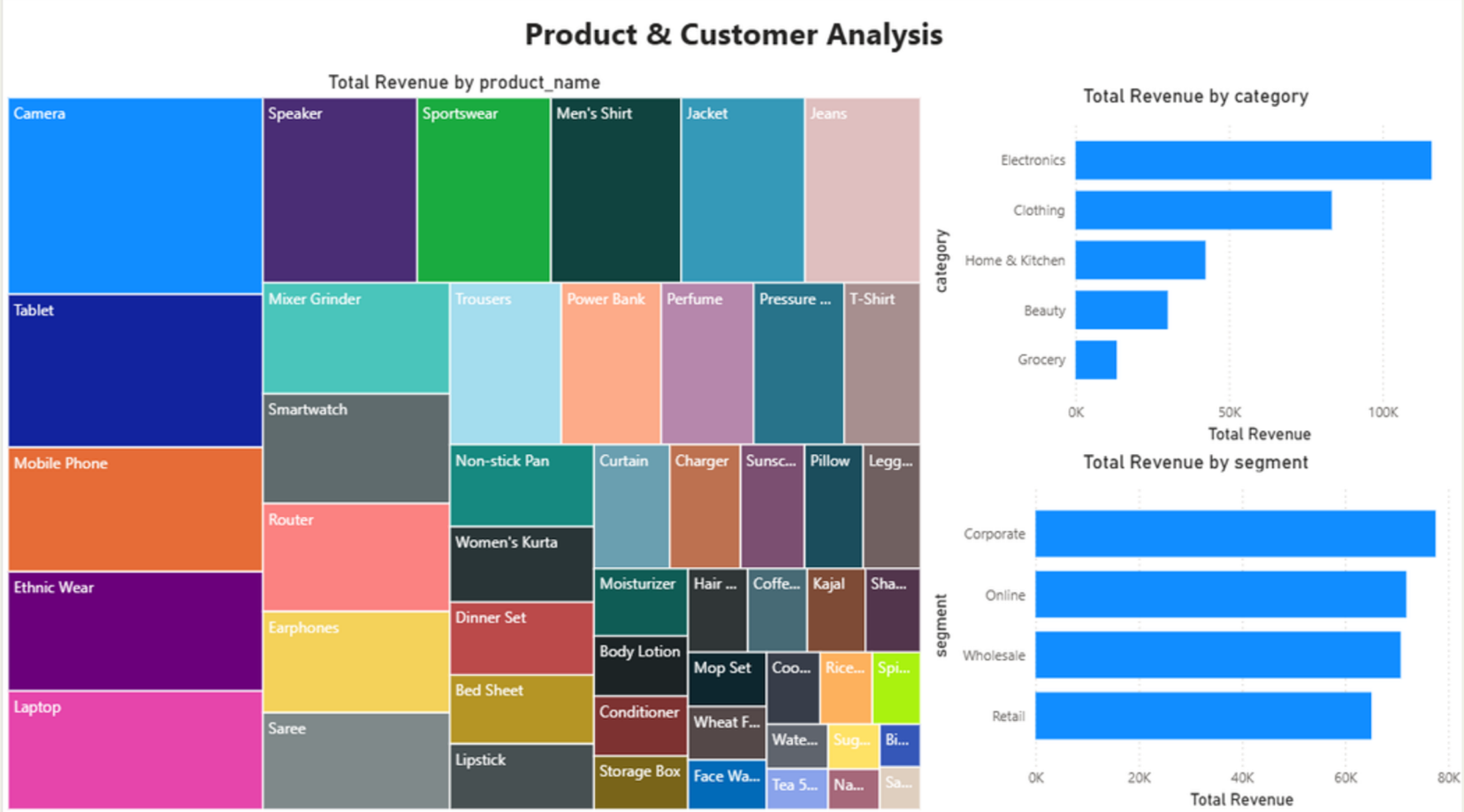
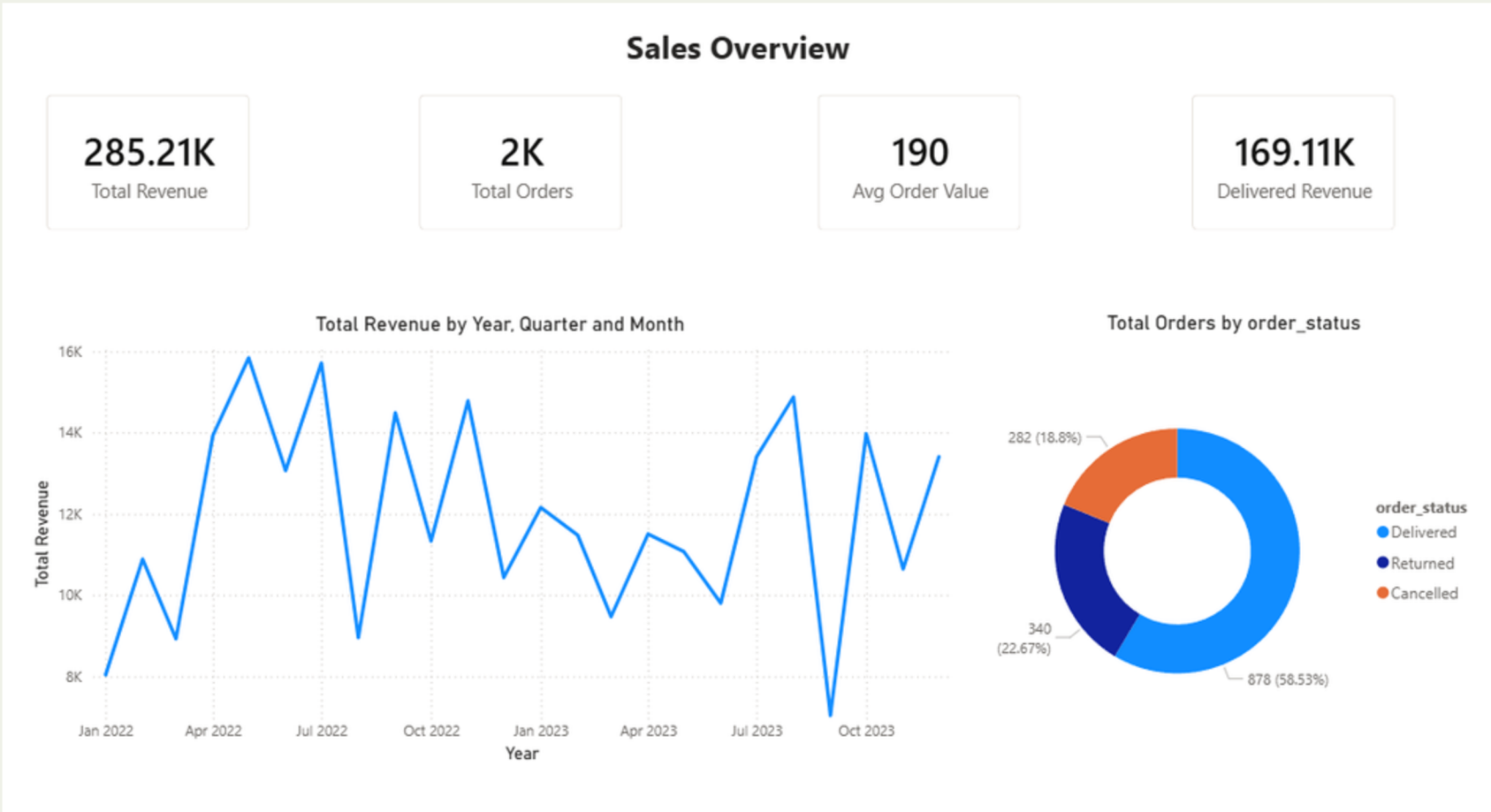
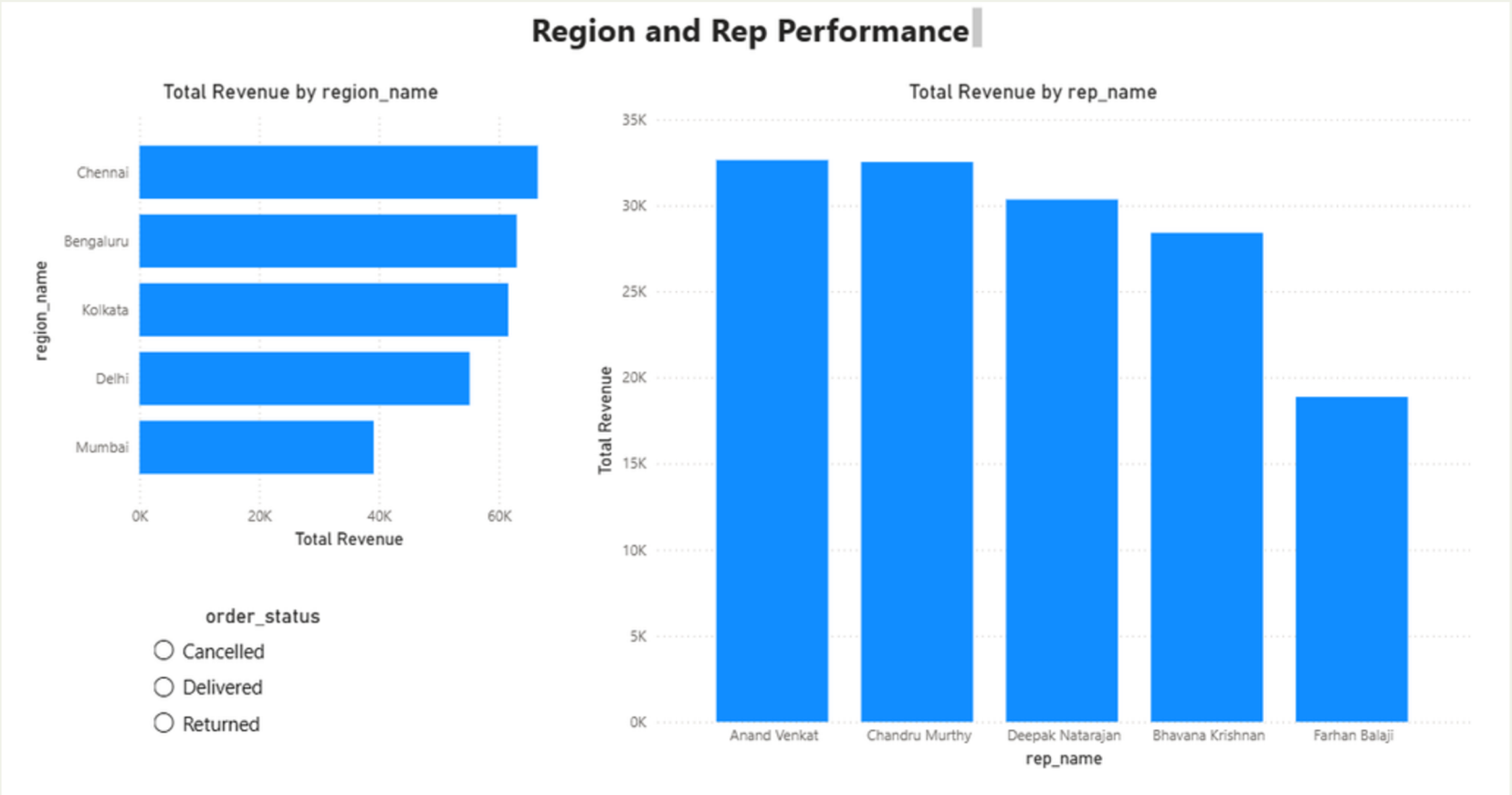
| Row Labels  | Count of sale_id | Sum of total_amount |
|-------------|------------------|---------------------|
| 0           | 649              | 125220.96           |
| 0.05        | 228              | 45363.8             |
| 0.1         | 198              | 37733.27            |
| 0.15        | 232              | 47077.5             |
| 0.2         | 193              | 29812.97            |
| Grand Total | 1500             | 285208.5            |

- **Discount Impact:** **649** orders with zero discount generated the highest revenue of **₹1,25,220**
- **Highest Discount Lowest Return:** Orders with **20%** discount generated the lowest revenue at **₹29,812**
- **Conclusion:** Higher discounts are not driving more sales or more revenue

# Interactive Dashboard — Power BI

3 page interactive dashboard connecting directly to MySQL database

- **Page 1:** Sales Overview - KPI cards, monthly trend, order status split
- **Page 2:** Region & Rep Performance - region revenue, top 5 reps
- **Page 3:** Product & Customer Analysis - category revenue, product treemap, segment analysis
- Dynamic slicers added for order status and category filtering across all pages



# Business Recommendations

## Problem 1 - Ineffective Discount Strategy

- **Customer View:** Customers are not motivated by discounts - they buy based on need and product value
- **Company View:** Blanket discounts are cutting into margins without driving revenue growth
- **Recommendation:** Stop blanket discounts - offer personalized discounts only to At Risk and Lost customers to win them back

## Problem 2 - Poor Order Conversion (41% incomplete)

- **Customer View:** Customers are dissatisfied with product quality or delivery experience - leading to returns and cancellations
- **Company View:** Every returned and cancelled order is a direct revenue loss and increases operational costs
- **Recommendation:** Investigate return reasons, improve product quality checks, strengthen delivery and after sales experience



# Business Recommendations

## Problem 3 - Poor Customer Retention

- Customer View: 47 At Risk customers are losing interest - they feel no reason to come back
- Company View: Acquiring new customers costs more than retaining existing ones - losing VIP and Loyal customers is a serious threat
- Recommendation: Launch a VIP loyalty program to retain top customers, run a re-engagement campaign for At Risk and Lost segments identified in RFM analysis

## Additional Recommendation:

- **Focus on Mumbai** - it is the lowest performing region across all categories, needs dedicated sales push and targeted marketing
- **Address September dip** - revenue crashes every September, plan promotional campaigns in advance to counter seasonal decline